

Week 6

# Personalized Federated Learning and Evaluation

When one global model is not enough

Graduate course · English instruction



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# Why Personalize?

A global model can capture shared structure, yet still perform poorly for clients whose local distributions differ from the population average.

## Global benefit

Shared training can improve generalization and data efficiency.

## Local mismatch

A single model may underperform on clients with distinct distributions or preferences.

## Personalization goal

Combine shared knowledge with client-specific adaptation.

Shared representation



client-specific adaptation



better local utility

Personalization is useful when clients are related—but not identical.

# Learning Objectives

## Motivation

Explain when a single global model is insufficient.

## Methods

Compare fine-tuning, layer separation, regularization, clustering, and mixture approaches.

## Representative algorithms

Understand the intuition behind FedPer, pFedMe, and related personalized FL families.

## Evaluation

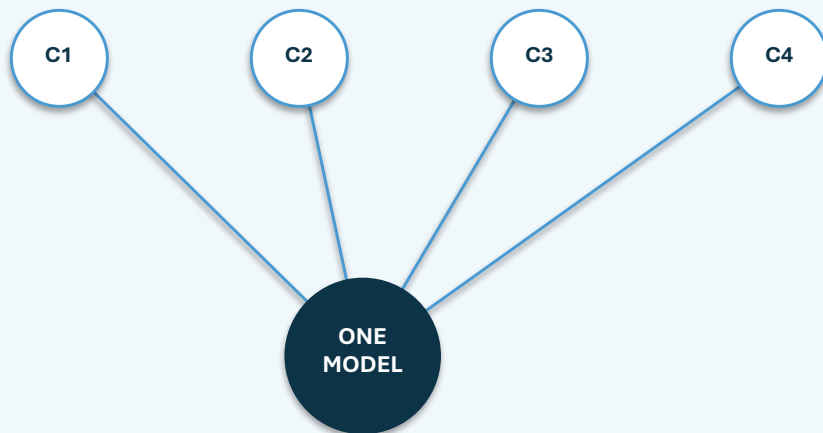
Measure global performance, client-level performance, fairness, and personalization gain.

**The goal is not simply to improve the mean—personalization should help the right clients for the right reason.**

# Global Model vs. Local Model

## Global-only model

Learns from many clients and can generalize broadly.

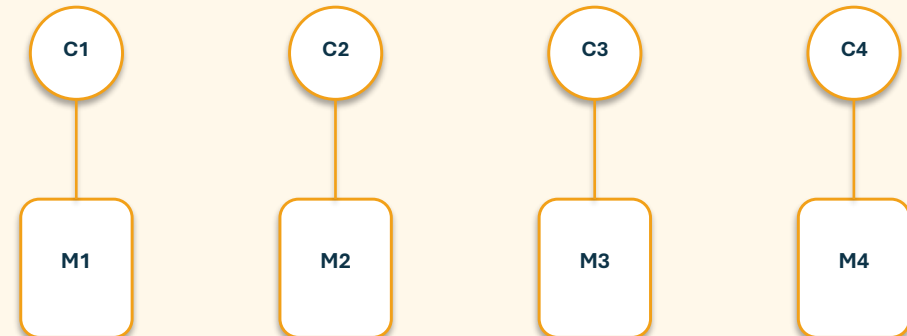


broad sharing

may ignore client-specific needs

## Local-only model

Matches one client distribution, but may overfit when local data are small or narrow.



high client specificity

less shared knowledge

# The Personalization Spectrum

Personalization is a spectrum—not a binary choice.

## Mostly shared

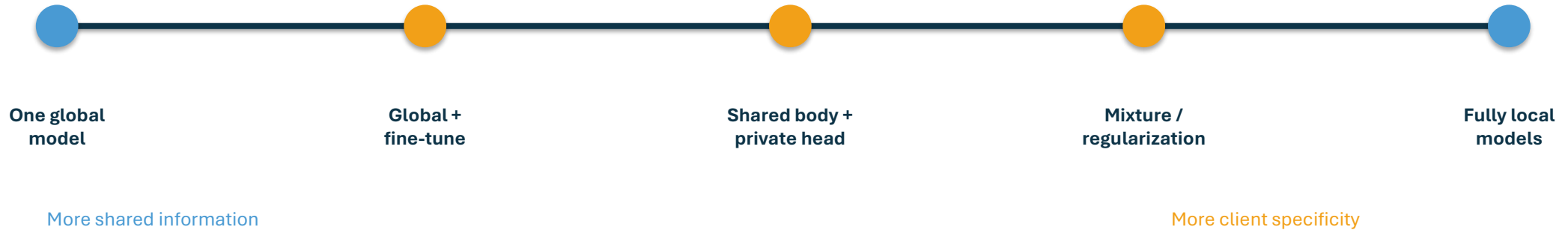
One global model maximizes knowledge sharing but may miss client-specific needs.

## Partially personalized

Fine-tuning, FedPer, and regularized methods share some knowledge while adapting locally.

## Mostly local

Fully local models maximize specificity but give up most cross-client knowledge transfer.



Methods differ in what is shared, what remains private, and when adaptation happens.

# A Useful Taxonomy of Personalized FL

## Fine-tuning

Adapt a learned global model locally after or during FL.

## Layer separation

Share some layers and keep client-specific parameters private.

## Regularization

Keep personalized models near a shared global reference.

## Clustering

Learn one model per group of similar clients.

## Mixture models

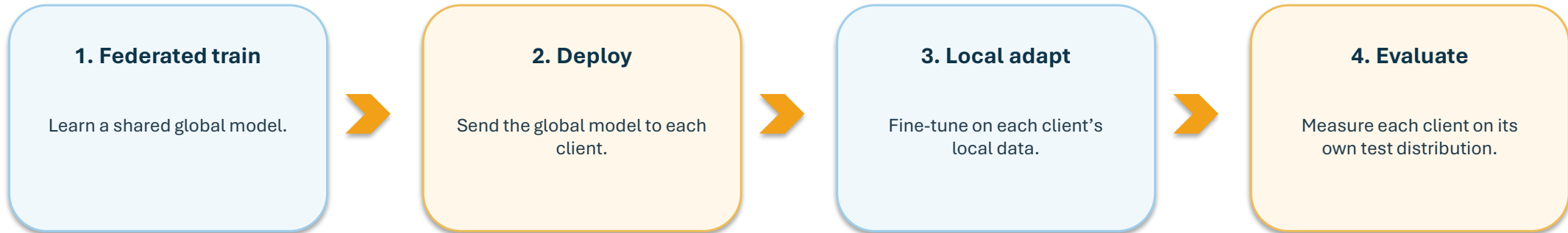
Combine local and global predictors.

## Multi-task / meta-learning

Treat clients as related tasks and learn fast adaptation.

**A useful taxonomy asks: which parameters are shared, how clients are related, and how personalization is enforced.**

# Fine-Tuning: The Strong Baseline



## Why it matters

Fine-tuning is simple, strong, and should usually be included as a personalization baseline.

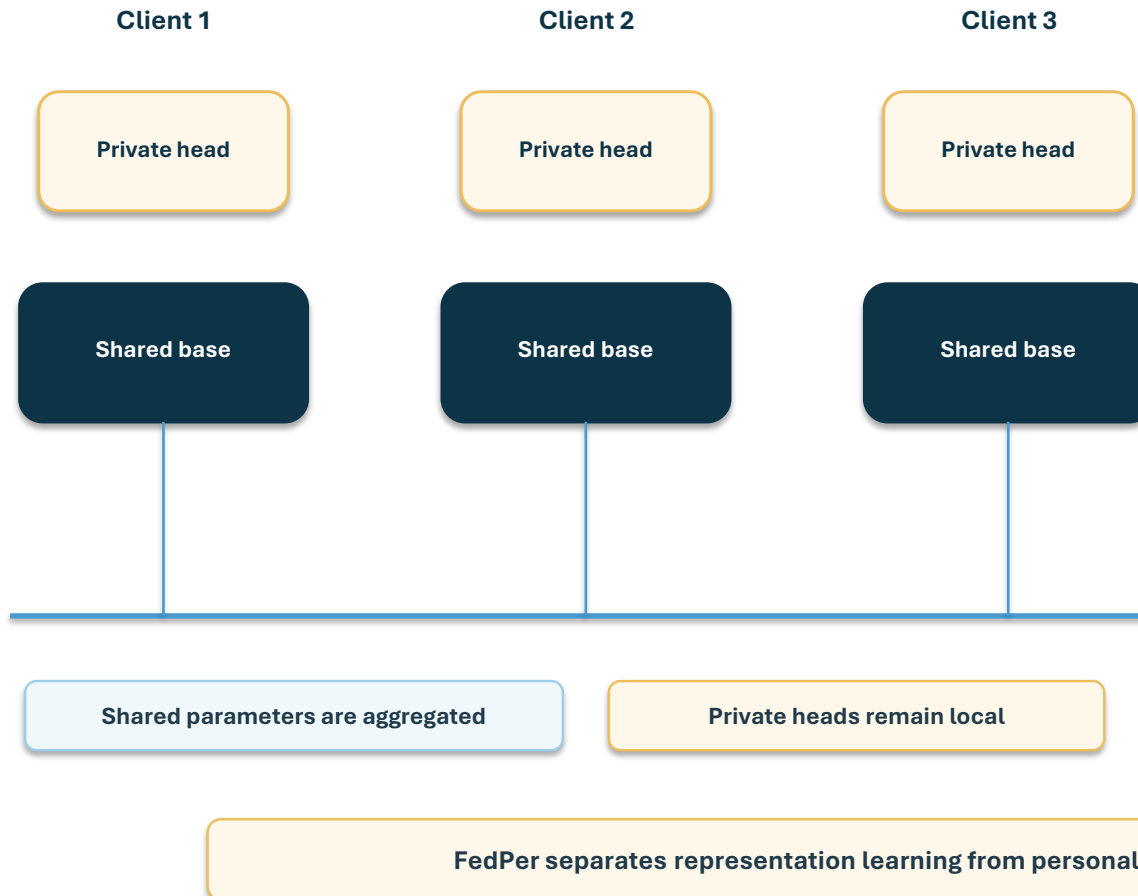
## Main risk

If local data are very small, aggressive fine-tuning may overfit or forget useful shared structure.

**Personalization gains should be compared against this simple baseline—not only against a global model.**

# Layer-Based Personalization: FedPer Intuition

FedPer-style parameter split: shared representation + client-specific prediction head



## What is shared?

The representation layers learn features from many clients and are aggregated by the server.

## What stays local?

Each client keeps its prediction head private and adapts it to its own local distribution.

## Why can this help?

Clients benefit from shared features while preserving client-specific decision boundaries.



# pFedMe: Regularize Personalized Models Toward a Global Model

$$F_k(\theta) = \min_{\theta_k} [ L_k(\theta_k) + (\lambda/2) \|\theta_k - \theta\|^2 ]$$

Each client optimizes a personalized model  $\theta_k$ , while  $\lambda$  controls how strongly it remains coupled to the global reference  $\theta$ .

## Small $\lambda$

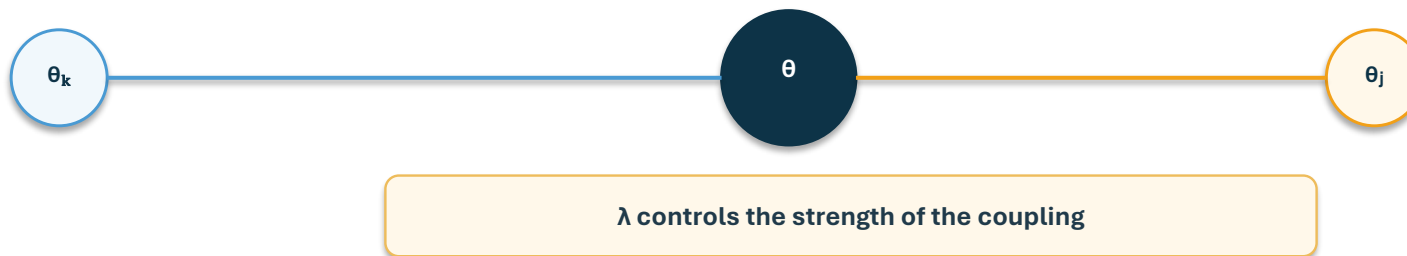
More local freedom.

## Large $\lambda$

Stronger global coupling.

## Key idea

Personalization becomes an explicit optimization objective.



# Other Personalized FL Families

## Client clustering

Find groups of similar clients and learn cluster-specific models.

## Meta-learning

Learn a global initialization that adapts quickly with few local steps.

## Mixture of models

Mix global and/or cluster predictors with client-specific weights.

## Multi-task learning

Jointly learn related client-specific tasks.

## Contextualization

Use side information or client context to guide personalization.

## Distillation

Transfer knowledge without requiring identical model architectures.

**Different families make different assumptions about how clients are related.**

# Research Example: Hierarchical Personalized FL with Selective Module Exchange

- **Local personalization:** clients keep private local models.
- **Selective sharing:** only prompt/adaptor updates are exchanged.
- **Hierarchical coordination:** client → edge → cloud aggregation.

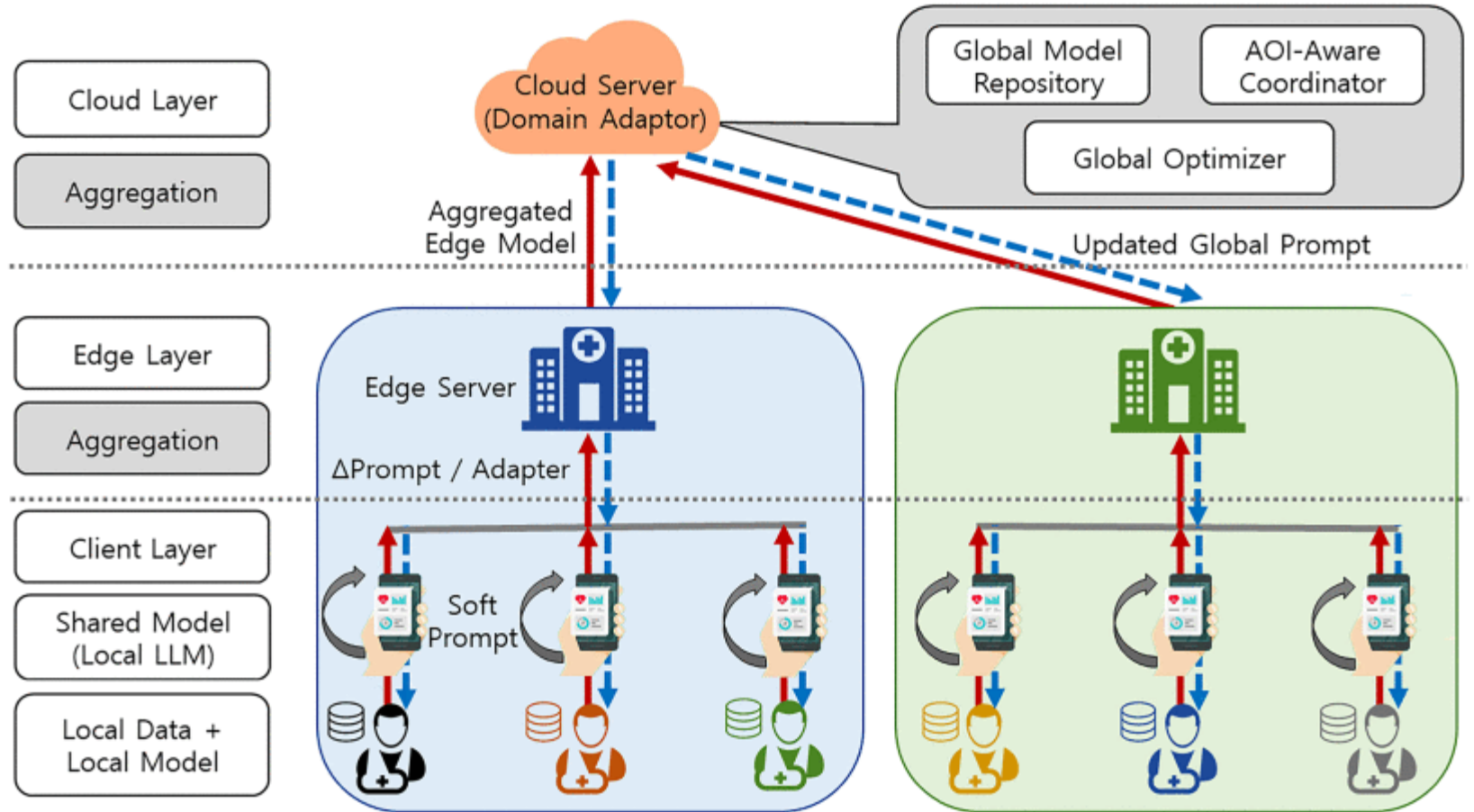


Fig. 1. Overview of the system architecture of the proposed EA-HPFL framework integrating Aol-aware coordination and prompt-based personalization across hierarchical edge–cloud layers.

# How Should Personalized FL Be Evaluated?

| Metric                  | Question answered                    | Why it matters               |
|-------------------------|--------------------------------------|------------------------------|
| Global accuracy         | How good is one shared model?        | Baseline                     |
| Mean client accuracy    | How does the typical client perform? | Client-level utility         |
| Worst / low-percentile  | Who is still underserved?            | Fairness and reliability     |
| Personalization gain    | Personalized – global per client     | Direct benefit of adaptation |
| Variance across clients | How unequal are outcomes?            | Stability and equity         |

**A personalized FL paper should report who benefits—not only whether the average improves.**

# Sample-Weighted vs. Client-Weighted Evaluation

## Sample-weighted

Larger clients contribute more to the final metric.



Reflects total examples

Can hide small-client failures

## Client-weighted

Every client contributes equally, regardless of dataset size.



Reflects typical client

More sensitive to small clients

The right weighting depends on the question the evaluation is meant to answer.

# Common Personalized FL Benchmarks

## FEMNIST

Partition unit

Writer

Task

Image classification

## Shakespeare

Partition unit

Speaking character

Task

Language modeling

## Reddit

Partition unit

User

Data

Large-scale language data

## Sent140

Partition unit

User

Task

Sentiment classification

**Benchmark choice should match the kind of client variation and personalization setting your method is designed to study.**

# In-Class Evaluation Exercise

## Scenario

Five clients obtain global accuracies [90, 88, 87, 62, 60] and personalized accuracies [91, 89, 88, 78, 75].

## Task 1

Compute the per-client personalization gain.

## Task 2

Would global average accuracy alone reveal the main benefit?

## Task 3

Which metric best supports a claim that personalization helps disadvantaged clients?

**Look at the client-level changes before computing any single summary number.**

# Key Takeaways

1

**Personalization balances federation-wide knowledge with client-specific adaptation.**

2

Fine-tuning is a strong baseline and should be included in comparisons.

3

FedPer separates shared and private parameters; pFedMe formulates personalization through regularized local objectives.

4

**Evaluation must be client-aware: average accuracy alone can hide who benefits and who is harmed.**

5

Next week: students critically present and discuss recent research papers from Weeks 1–6.

**Mental model: decide what should be shared → decide what should be personalized → evaluate each client.**



# Reading & Preparation

## Main textbook

Ludwig & Baracaldo (2022), Chapters 4–5 on personalization and robust personalized FL.

## Supplementary reference

Kairouz et al. (2021), Section 3.3: Multi-Task Learning, Personalization, and Meta-Learning.

## Prepare for Week 7 — Student Paper Presentations I

- 1 Prepare one recent federated learning paper from the approved topic area.
- 2 Focus on problem formulation, assumptions, method, experiments, and limitations.
- 3 End with a clear research gap or possible extension.

**For Week 7, explain the paper as a researcher—not as a slide reader.**

Q & A

Thank You.

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